

James D. Motes
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RESEARCH PROFILE

I am a robotics researcher developing planning systems that enable robot teams to work together on complex tasks. My work makes coordination practical as teams grow by identifying which decisions require joint planning and accelerating the necessary computation. I am extending these foundations through learning and human-robot interaction, so teams can learn from experience and people can guide and revise their plans.

ACADEMIC APPOINTMENT

Postdoctoral Researcher, University of Illinois Urbana-Champaign
Advisor: Dr. Nancy M. Amato

August 2023–Present

EDUCATION

Ph.D. in Computer Science, University of Illinois Urbana-Champaign
Ph.D. Thesis: Multi-robot Task and Motion Planning in Hybrid State Spaces
Advisor: Dr. Nancy M. Amato

August 2023

M.S. in Computer Science, Texas A&M University
M.S. Thesis: Interaction Templates for Multi-Robot Systems
Advisor: Dr. Nancy M. Amato

August 2019

B.S. in Computer Engineering, Texas A&M University
Minor in Mathematics, Engineering Honors, Undergraduate Research Scholar

May 2018

RESEARCH LEADERSHIP

- **Research direction.** Currently lead research teams involving 12+ graduate students in the Parasol Lab, setting project directions and coordinating research and publication development.
- **Graduate mentorship.** Mentored 20+ graduate students across doctoral and postdoctoral work, including primary day-to-day mentorship of two Ph.D. and three M.S. graduates. Mentorship contributed to 11 student-led archival publications.
- **Open-source research software.** Primary developer and architectural lead of [CoMotion](#); direct graduate-student development of [Open-SPITE](#) and coordinate development of the [Parasol Planning Library](#).
- **Proposals and collaborations.** Primary author of an NSF robotics proposal (pending) and eight industry proposals; coordinate interdisciplinary and industry research collaborations.

PUBLICATIONS & SCHOLARLY OUTPUTS

Publication Overview

- 16 peer-reviewed archival publications, including 12 journal articles and 4 refereed conference proceedings.
- 4 first-author peer-reviewed publications and 2 first-author manuscripts under review.
- 11 student-led archival publications directly mentored, with additional manuscripts under review.
- Additional scholarly outputs include preprints, technical reports, workshop papers, and extended abstracts.
- [Google Scholar](#): 415+ citations, h-index: 7, as of September 26, 2026.

Notation

- An underlined name indicates a student for whom Motes served as primary day-to-day research mentor.
- * indicates equal contribution.

Peer-Reviewed Archival Publications

Journal Articles

- [J12] J.D. Motes, M. Morales, and N. M. Amato, “P-ARC: Exploiting Subproblem Independence for Parallel Multi-Robot Motion Planning”, *IEEE Robotics and Automation Letters (RA-L)*, 2026.
- [J11] S. Lee, J.D. Motes, I. Ngui, M. Morales, and N. M. Amato, “Lazy-DaSH: A Lazy Approach for Hypergraph-Based Multirobot Task and Motion Planning”, *IEEE Transactions on Robotics (T-RO)*, vol. 42, 2026.
- [J10] C. McBeth, J.D. Motes, I. Ngui, M. Morales, and N. M. Amato, “Scalable Multi-Robot Motion Planning Using Workspace Guidance-Informed Hypergraphs”, *IEEE Robotics and Automation Letters (RA-L)*, vol. 11, 2026.

- [J9] [H. Lee](#), J.D. Motes, M. Morales, and N. M. Amato, “An Analysis of Constraint-Based Multiagent Pathfinding Algorithms”, *IEEE Transactions on Robotics (T-RO)*, vol. 42, 2026.
 - Presented at *IEEE International Conference on Robotics and Automation (ICRA)*, Vienna, Austria, June 2026.
- [J8] [M. Qin](#), [I. Solis](#), J.D. Motes, M. Morales, and N. M. Amato, “K-ARC: Adaptive Robot Coordination for Multi-Robot Kinodynamic Planning”, *IEEE Robotics and Automation Letters (RA-L)*, vol. 10, 2025.
 - Presented at *IEEE International Conference on Robotics and Automation (ICRA)*, Vienna, Austria, June 2026.
- [J7] [I. Solis](#), J.D. Motes, [M. Qin](#), M. Morales, and N. M. Amato, “Adaptive Robot Coordination: A Subproblem-based Approach for Hybrid Multi-Robot Motion Planning”, *IEEE Robotics and Automation Letters (RA-L)*, 2024.
 - Presented at *IEEE ICRA@40*, Rotterdam, The Netherlands, Sept. 2024.
- [J6] [C. McBeth](#), J.D. Motes, D. Uwacu, M. Morales, and N. M. Amato, “Scalable Multi-Robot Motion Planning for Congested Environments With Topological Guidance”, *IEEE Robotics and Automation Letters (RA-L)*, 2023.
 - Presented at *IEEE International Conference on Robotics and Automation (ICRA)*, Yokohama, Japan, May 2024.
- [J5] J.D. Motes, T. Chen, T. Bretl, M. Morales, and N. M. Amato, “Hypergraph-Based Multi-Robot Task and Motion Planning”, *IEEE Transactions on Robotics (T-RO)*, 2023.
 - Presented at *IEEE International Conference on Robotics and Automation (ICRA)*, Yokohama, Japan, May 2024.
- [J4] [H. Lee](#), J.D. Motes, M. Morales, and N. M. Amato, “Parallel Hierarchical Composition Conflict-Based Search for Optimal Multi-Agent Pathfinding”, *IEEE Robotics and Automation Letters (RA-L)*, vol. 6, 2021.
 - Presented at *IEEE International Conference on Intelligent Robots and Systems (IROS)*, Prague, Czech Republic, 2021 (virtual).
- [J3] [I. Solis](#), J.D. Motes, R. Sandström, and N. M. Amato, “Representation-Optimal Multi-Robot Motion Planning Using Conflict-Based Search”, *IEEE Robotics and Automation Letters (RA-L)*, vol. 6, 2021.
 - Presented at *IEEE International Conference on Robotics and Automation (ICRA)*, Xi’an, China, 2021 (virtual).
- [J2] J.D. Motes, R. Sandström, [H. Lee](#), S. Thomas, and N. M. Amato, “Multi-Robot Task and Motion Planning With Subtask Dependencies”, *IEEE Robotics and Automation Letters (RA-L)*, vol. 5, 2020.
 - Presented at *IEEE International Conference on Robotics and Automation (ICRA)*, Paris, France, 2020 (virtual).
- [J1] J.D. Motes, R. Sandström, W. Adams, [T. Ogunyale](#), S. Thomas, and N. M. Amato, “Interaction Templates for Multi-Robot Systems”, *IEEE Robotics and Automation Letters (RA-L)*, vol. 4, 2019.
 - Presented at *IEEE International Conference on Intelligent Robots and Systems (IROS)*, Macao, China, 2019.

Refereed Conference Proceedings

- [C4] [I. Ngui](#), [C. McBeth](#), A. Santos, G. He, K. J. Mimnaugh, J.D. Motes, L. Soares, M. Morales, and N. M. Amato, “ERUPT: An Open Toolkit for Interfacing with Robot Motion Planners in Extended Reality”, in *International Conference on Artificial Reality and Telexistence and Eurographics Symposium on Virtual Environments (ICAT-EGVE)*, 2026.
 - Presented at *ICAT-EGVE 2026*, Goiânia, Brazil, Sept. 2026.
- [C3] [I. Ngui](#), [C. McBeth](#), J.D. Motes, M. Morales, and N. M. Amato, “Scalable Multi-Robot Motion Planning via Hierarchical Subproblem Expansion and Workspace Decomposition Refinement”, in *Algorithmic Foundations of Robotics XVII: Proc. the Seventeenth World Symposium on the Algorithmic Foundations of Robotics (WAFR)*, 2026.
- [C2] [S. Ashur](#), M. Lusardi, [M. Markowicz](#), J.D. Motes, M. Morales, S. Har-Peled, and N. M. Amato, “SPITE: Simple Polyhedral Intersection Techniques for Motion Planning in Modified Environments”, in *Algorithmic Foundations of Robotics XVI, Volume 1*, Springer Proceedings in Advanced Robotics, vol. Vol. 37, Springer, 2026.
 - Presented at *Workshop on the Algorithmic Foundations of Robotics (WAFR)*, 2024.

- [C1] A. Goddu, T. Brown, M. Paton, J.D. Motes, and T. Chen, “Modifying ABIT* for Tethered Rappelling Robot Motion Planning”, in *Proc. 2025 IEEE 21st International Conference on Automation Science and Engineering (CASE)*, 2025.

Manuscripts Under Review

First Author

- [U1] J.D. Motes, M. Morales, and N. M. Amato, “Multi-Robot Motions in Milliseconds: Vector-Accelerated Primitives for Sampling-Based Planning”, *arXiv preprint arXiv:2604.23960*, 2026.
- [U2] J.D. Motes, M. Morales, and N. M. Amato, “AO-ARC: Almost-Surely Asymptotically Optimal Multi-Robot Motion Planning with ARC”, *arXiv preprint arXiv:2606.27495*, 2026.

Mentored Student-Led Manuscripts

- [U6] V. Murta, I. B. Love, J.D. Motes, M. Morales, and N. M. Amato, “RoboCheck: A Fast, Exact Massively Parallel Collision Checker for Mobile Robotics”, submitted, 2026.
- [U5] M. Markowicz, J.D. Motes, M. Morales, and N. M. Amato, “MR-SPITE: Accelerating Multi-Robot Conflict Scans via Hierarchical Swept-Volume Approximations”, submitted, *arXiv preprint arXiv:2609.23928*, 2026.
- [U4] Y. Arad, M. Markowicz, S. Ashur, J.D. Motes, M. Morales, and N. M. Amato, “Serialized Red-Green-Gray: Quicker Heuristic Validation of Edges in Dynamic Roadmap Graphs”, *arXiv preprint arXiv:2603.28674*, 2026.
- [U3] I. Solis, J.D. Motes, J. Montalvo, M. Qin, M. Morales, and N. M. Amato, “Experience-Based Subproblem Planning for Multi-Robot Motion Planning”, *arXiv preprint arXiv:2411.08851*, 2024.

Additional Scholarly Outputs

Preprints and Technical Reports

- [TR2] H. Lee, Z. Serlin, J.D. Motes, B. Long, M. Morales, and N. M. Amato, “PRISM: Complete Online Decentralized Multi-Agent Pathfinding With Rapid Information Sharing Using Motion Constraints”, *arXiv preprint arXiv:2505.08025*, 2025.
- [TR1] A. Attali, S. Ashur, I. B. Love, C. McBeth, J.D. Motes, D. Uwacu, M. Morales, and N. M. Amato, “A Framework for Guided Motion Planning”, *arXiv preprint arXiv:2404.03133*, 2024.

Workshop Papers and Extended Abstracts

- [W9] M. Markowicz, S. Ashur, J.D. Motes, and N. M. Amato, “Semi-Lazy Rearrangement Solver for Accelerating Task Planning Using SPITE Dynamic Roadmaps”, in *Extended Abstract, 4th Workshop on Future of Construction: Safe, Reliable, and Precise Robots in Construction Environments (ICRA 2025)*, Atlanta, GA, USA, May 19, 2025.
- [W8] H. Lee, J.D. Motes, Z. Serlin, M. Morales, and N. M. Amato, “Distributed Constraint-Based Search Using Multi-Hop Communication”, in *Extended Abstract, ICRA@40: 40th Anniversary of the IEEE Conference on Robotics and Automation (ICRA@40)*, Rotterdam, The Netherlands, Sept. 23–26, 2024.
- [W7] I. Ngui, J.D. Motes, H. Lee, M. Morales, and N. M. Amato, “Ergonomic Motion Planning for Human-Robot Collaboration”, in *Extended Abstract, ICRA@40: 40th Anniversary of the IEEE Conference on Robotics and Automation (ICRA@40)*, Rotterdam, The Netherlands, Sept. 23–26, 2024.
- [W6] K. Elimelech*, J.D. Motes*, M. Morales, N. M. Amato, M. Vardi, and L. E. Kavraki, “Encoding Reusable Multi-Robot Planning Strategies as Abstract Hypergraphs”, in *Extended Abstract, ICRA@40: 40th Anniversary of the IEEE Conference on Robotics and Automation (ICRA@40)*, Rotterdam, The Netherlands, Sept. 23–26, 2024.
- [W5] S. Lee, J.D. Motes, I. Ngui, M. Morales, and N. M. Amato, “Lazy-DaSH: Lazy Approach for Hypergraph-based Multi-Robot Task and Motion Planning”, in *Extended Abstract, ICRA@40: 40th Anniversary of the IEEE Conference on Robotics and Automation (ICRA@40)*, Rotterdam, The Netherlands, Sept. 23–26, 2024.
- [W4] M. Lusardi*, M. Markowicz*, S. Ashur, J.D. Motes, M. Morales, S. Har-Peled, and N. M. Amato, “SPITE: Simple Polyhedral Intersection Techniques for Modified Environments”, in *Extended Abstract, ICRA@40: 40th Anniversary of the IEEE Conference on Robotics and Automation (ICRA@40)*, Rotterdam, The Netherlands, Sept. 23–26, 2024.

- [W3] I. Ngui, S. Lee, J.D. Motes, M. Morales, and N. M. Amato, “A Hierarchical Approach to Workstation-based Task-Allocation and Motion Planning”, in *Extended Abstract, RAFF2023: Robotics & AI in Future Factory (IROS 2023 Workshop)*, IROS 2023, Detroit, MI, USA, Oct. 1–5, 2023.
- [W2] A. Attali, S. Ashur, I. B. Love, C. McBeth, J.D. Motes, D. Uwacu, M. Morales, and N. M. Amato, “Evaluating Guiding Spaces for Motion Planning”, in *Extended Abstract, Evaluating Motion Planning Performance, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2022)*, Kyoto, Japan, Oct. 23, 2022.
- [W1] T. Chen, Z. Huang, J.D. Motes, J. Geng, Q. Ta, H. Dinkel, H. Abdul-Rashid, J. Myers, Y. Mun, W. Lin, Y. Huang, S. Liu, M. Morales, N. M. Amato, K. Driggs-Campbell, and T. Bretl, “Insights from an Industrial Collaborative Assembly Project: Lessons in Research and Collaboration”, in *IEEE ICRA Workshop on Collaborative Robots and the Work of the Future*, Philadelphia, USA, 2022.

OPEN-SOURCE RESEARCH SOFTWARE

CoMotion — Coordinated Motion Planning for Multi-Robot Systems

Research Software Lead

Open-source C++ platform integrating selective-coordination multi-robot planning with accelerated planning infrastructure, including ARC-family methods, parallel planning, and accelerated motion validation. Primary developer of the library’s code; lead its architecture, technical direction, and ongoing development.

Open-SPITE

Research Software Lead

Open-source C++ library for efficient motion validation and replanning in changing environments, including SPITE and Serialized Red-Green-Gray (SerRGG). Designed the architecture and development plan; direct and coordinate graduate-student implementation, integration, and maintenance.

Parasol Planning Library (PPL)

Research Software Lead

Shared C++ robotics planning infrastructure supporting Parasol Lab research. Previously the library’s most prolific contributing developer; current responsibilities focus on managing the codebase and coordinating development.

PROPOSAL & SPONSORED RESEARCH DEVELOPMENT

National Science Foundation (NSF) – Foundational Research in Robotics (FRR) Proposal

September 2025

- Primary author of proposal to customize robotic motion behaviors for non-expert users through natural language preference specification (Status: Pending).
- Coordinated collaboration with natural language processing researchers at the University of Illinois Urbana-Champaign.
- Developed research narrative connecting human preference specification, motion planning, and human-steerable robot behavior.

National Science Foundation (NSF) – Engineering Research Center (ERC) Site Visit

March 2026

- Supported preparation for an NSF Engineering Research Center site visit by developing research-thrust narrative and presentation materials.

Industry Proposals and White Papers

April 2025–Present

- Primary author of eight proposals and secondary author on four more.
- Proposal topics include human-robot interaction, multi-robot planning, environment design, and data-driven planning.
- Developed sponsored research narratives connecting scalable robot-team planning to deployable industrial automation systems.

INVITED TALKS & SEMINARS

Adaptive Robot Coordination: Lessons, Limitations, and Opportunities

- Aggie Autonomous Robotics Research Group, Department of Computer Science and Engineering, Texas A&M University, College Station, TX.
- August 2026. Invited lab seminar.

Lessons from Manufacturing Startups

- AI Meets Robots & Machines Seminar, University of Illinois Urbana-Champaign, Urbana, IL.
- February 2026. Invited seminar.

Multi-Robot Task and Motion Planning in Hybrid State Spaces

- Kavradi Lab, Department of Computer Science, Rice University, Houston, TX.
- February 2024. Invited lab Seminar.

Planning Motions and Tasks for Manipulators, Multi-Robot Systems and Biomolecules

- Center for Autonomy Portfolio Discovery Workshop, University of Illinois Urbana-Champaign, Urbana, IL.
- October 2019. Invited Workshop Talk.

TEACHING & CURRICULUM DEVELOPMENT

Lead Instructor, AI4ALL

August 2021–December 2023

- Instructor for introductory artificial intelligence courses delivered through partner programs at the University of Texas at El Paso, New Mexico State University, Worcester Polytechnic Institute, and Texas A&M University.
- Developed curriculum for the Discover AI introductory artificial intelligence course.
- Led and mentored instructor teams across participating universities.
- Teaching assistant for Discover AI at the University of Illinois Urbana-Champaign in Fall 2021.

Teaching Assistant, University of Illinois Urbana-Champaign

January 2021–May 2021

- Teaching assistant for Computer Science Ph.D. seminar course.
- Organized guest speakers, managed grading, and coordinated student assignments.

Internal Lab Instruction, Parasol Lab

August 2019–May 2021

- Developed and taught a motion-planning course for new Parasol Lab members.

MENTORING, RESEARCH SUPERVISION & OUTREACH

Graduate Research Leadership & Mentorship, Parasol Lab

- Mentored 20+ graduate students across doctoral and postdoctoral work; currently lead research teams involving 12+ graduate students.
- Set project directions and coordinate research milestones, paper development, and shared software integration. Mentorship contributed to 11 student-led archival publications.

Selected Completed Graduate Research Projects

Served as the primary day-to-day research mentor throughout the following projects.

- **Irving Solis, Ph.D. (2024)** — Adaptive Robot Coordination (ARC) and multi-robot motion planning.
- **Hannah Lee, Ph.D. (2024)** — Multi-agent path finding; parallel and task-and-motion variants of conflict-based search.
- **Mike Qin, M.S. (2024)** — Kinodynamic Adaptive Robot Coordination (K-ARC) and multi-robot kinodynamic planning.
- **Yulie Arad, M.S. (2026)** — Dynamic-roadmap validation and replanning, including SPITE and Serialized Red-Green-Gray (SerGG).
- **Jared Buabeng, M.S. (2026)** — ORBIT and language-conditioned motion planning.

Undergraduate Research Mentorship

- During graduate study, mentored undergraduate researchers, including local university students and DREU program participants.

Houston Robotics Club

August 2023–Present

- Mentor community college and high school students on AI and robotics projects.
- Supported students who later enrolled in STEM undergraduate programs or obtained internships with local robotics companies.

PROFESSIONAL SERVICE

Program Committee Member

- World Symposium on Algorithmic Foundations of Robotics, 2026.
- World Symposium on Algorithmic Foundations of Robotics, 2024.

Reviewer/Referee

August 2018–Present

- IEEE Robotics and Automation Letters (RA-L)
- IEEE Transactions on Robotics (T-RO)
- IEEE International Conference on Robotics and Automation (ICRA)
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
- IEEE International Symposium on Multi-robot and Multi-agent Systems (MRS)
- World Symposium (formerly International Workshop) on Algorithmic Foundations of Robotics (WAFR)
- International Journal of Robotics Research (IJRR)
- Artificial Intelligence Journal (AIJ)

Professional Memberships

- IEEE Robotics and Automation Society

May 2019–Present

INDUSTRY, ENTREPRENEURSHIP & RESEARCH TRANSLATION

Optigon Inc., Principal Software Engineer, part time

May 2024–May 2026

- Led software development for DOE SBIR-funded high-throughput, non-contact spectroscopic metrology and analysis tools for clean-energy materials, including perovskite photovoltaics.
- Developed machine learning analytical models of solar cell behavior.

Normandy Automation, Founder

May 2023–August 2025

- Developed autonomous robotic manufacturing concepts for just-in-time manufacturing with minimal human reprogramming.
- Leveraged AI and multi-robot systems for autonomous welding and CNC machine-tending workflows.
- Wrote NSF SBIR proposals for autonomous robotic manufacturing systems.

Industrial Human-Robot Factory Collaboration, UIUC

August 2019–August 2023

- Collaborated with Foxconn Interconnect Technologies and UIUC research groups led by Timothy Bretl and Katherine Driggs-Campbell on collaborative human-robot factory projects.
- Connected multi-robot task and motion planning research to industrial assembly and human-robot collaboration settings.

HONORS

Engineering Graduate Merit Fellowship

August 2018–August 2019

Presidential Endowed Scholarship

August 2014–May 2018

Distinguished Student Award – Dwight Look College of Engineering

May 2017

Industrial Affiliates Program Scholarship

August 2014–May 2018

Dell Merit Scholarship

August 2017–May 2018